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The relative performance of AIC, AIC_C and BIC in the presence of unobserved heterogeneity

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Summary

1. Model selection is difficult. Even in the apparently straightforward case of choosing between standard linear regression models, there does not yet appear to be consensus in the statistical ecology literature as to the right approach.
2. We review recent works on model selection in ecology and subsequently focus on one aspect in particular: the use of the Akaike Information Criterion (AIC) or its small-sample equivalent, AIC_C. We create a novel framework for simulation studies and use this to study model selection from simulated data sets with a range of properties, which differ in terms of degree of unobserved heterogeneity. We use the results of the simulation study to suggest an approach for model selection based on ideas from information criteria but requiring simulation.
3. We find that the relative predictive performance of model selection by different information criteria is heavily dependent on the degree of unobserved heterogeneity between data sets. When heterogeneity is small, AIC or AIC_C are likely to perform well, but if heterogeneity is large, the Bayesian Information Criterion (BIC) will often perform better, due to the stronger penalty afforded.
4. Our conclusion is that the choice of information criterion (or more broadly, the strength of likelihood penalty) should ideally be based upon hypothesized (or estimated from previous data) properties of the population of data sets from which a given data set could have arisen. Relying on a single form of information criterion is unlikely to be universally successful.

Key-words: Akaike Information Criterion, Bayesian Information Criterion, generalized linear models, likelihood penalization, linear regression, model selection, statistical controversies

Introduction

The topic of model selection continues to receive considerable attention in statistical ecology. This work will present a brief review of the topic before examining the most widely used current approach, and developing a novel simulation framework for assessing the performance of a set of model selection strategies. We then suggest how to use this framework as a practical, data-based model selection strategy. The focus throughout will be on linear models and generalized linear models (GLMs), although the general principles extend to all classes of model; in this context, the term ‘model selection’ will be interchangeable with ‘variable selection’, although more generally the former term is more broad as it can involve choosing between different classes or types of models, for example.

REVIEW

Much current debate and discussion on model selection in ecology stems from the well-known text Burnham & Anderson

(2002), which recommended the Akaike Information Criterion (AIC; Akaike 1973) be used for model selection, ideally in a multimodel context. Theoretically, AIC is optimal in the sense of minimizing the mean square error of predictions, whereas a competing metric, the Bayesian Information Criterion (BIC; Schwarz 1978), is *consistent* – that is, it will tend to select the ‘true’ model as the sample size increases to infinity. The concept of a true model is only useful if: (a) it exists and contains only a finite number of explanatory variables; and (b) the full set of explanatory variables is available as data. If (b) is not satisfied, Burnham & Anderson (2002, §6.3.4) suggest BIC will in practice tend to find the ‘quasi-true’ model, one which contains (only) the subset of explanatory variables from the true model which *are* in the data set. A number of authors have argued that (a) is unlikely to ever be true in biological applications: for a recent discussion, see Aho, Derryberry & Peterson (2014). The standard argument (expressed also in Buckland, Burnham & Augustin 1997) is that the true model is too complex to model exactly (with number of parameters p greater than the sample size n), and that effect sizes will tend to taper: there will be a few large, important effects, followed by very many decreasing effects. This led Burnham & Anderson (2004) to

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state that one should use 'AIC for tapering effects and BIC for when there are no effects at all or a few big effects and all others are zero effects (no intermediate effects, no tapering effects)', although this is not a universally held view: Raffalovich *et al.* (2008) claimed via simulation experiments that BIC was generally preferable in all cases. We examine this issue closely in a simulation study.

AIC is defined as

$$\text{AIC} = -2 \log \ell(\hat{\theta}) + 2p \quad \text{eqn 1}$$

while BIC is defined as

$$\text{BIC} = -2 \log \ell(\hat{\theta}) + p \log n \quad \text{eqn 2}$$

where $\hat{\theta}$ represents the maximum likelihood estimates of the model parameters, $\log \ell(\hat{\theta})$ is the corresponding log-likelihood, and the final terms in both (1) and (2) represent a *penalty* on the log-likelihood as a function of the number of parameters p ; with linear models, these will mostly be regression coefficients. The penalties are there to reduce the effects of overfitting, and note that the penalty is stronger for BIC than AIC for any reasonable sample size. For small n , a corrected version of AIC (AIC_C) exists:

$$\text{AIC}_C = \text{AIC} + \frac{2p(p+1)}{n-p-1},$$

which provides a stronger penalty than AIC for smaller sample sizes, and stronger than BIC for very small sample sizes. Since AIC_C is reported to have better small-sample behaviour and since also $\text{AIC}_C \rightarrow \text{AIC}$ as $n \rightarrow \infty$, Burnham & Anderson (2002) recommended use of AIC_C as standard. The effect of a stronger penalty on the likelihood is to select smaller models, and so BIC tends to choose smaller models than AIC, and AIC_C also chooses relatively small models for smaller sample sizes. In consequence, Burnham & Anderson (2002, p. 33) claimed that BIC 'tend[s], in realistic situations, to select models that are too simple (i.e. underfitted)'. We examine this claim in a simulation study; in this paper, we focus on the relative prediction performance of the three information criteria, and relate results to the concept of strength of penalty. In particular, we ask: Are there occasions when models selected by BIC (or some other degree of penalty) would produce better predictions than models selected by either AIC or AIC_C ?

FINDING A 'BEST' MODEL

In their classic regression text, Draper & Smith (1966) included a section on model selection and called it *Selecting the 'Best' Regression Equation*; the quotation marks reflect the difficulty in determining what makes a fitted model 'best', and also how to find it; this has been a long-standing issue in statistics. In the third edition (Draper & Smith 1998), the section had become a whole chapter (15), with a section on the need to exercise caution when selecting models based on observational data. Potential problems include omitted variable bias ('latent or lurking variables'; Draper & Smith 1998) observed (p. 328) that an 'observed false effect of a visible variable' may be due

to an omitted variable, itself correlated with the visible variable. Since the values of the omitted variable are not part of the selected model, substantial changes between data sets can cause predictions from the model to 'become unreliable'.

This is an aspect of model selection which is typically ignored (but was recognized by Wenger & Olden 2012, for example). Theoretical justifications for AIC's optimality assume that the sampling regime remains constant across repeated data sets. Shibata (1981) proved the optimality of AIC with respect to '[t]he expectation, with respect to future observations, of the sum of squared errors of prediction', under the assumption for sample size n that 'the regression function is specified by infinitely many nonzero parameters or by an increasing number of variables as $n \rightarrow \infty$ '. An unwritten but necessary additional assumption is that the underlying regression model and data processes are constant across repeated sampling. It is this assumption which can be difficult to justify or verify in practice; we suspect that it is often not realistic. For example, it is common to attempt to use a model fitted at one sampling location (or 'site') to predict responses at another, and spatial variability may worsen predictive accuracy; temporal variation may also make prediction difficult at the same site but at different times. One question we might ask is: What effects do different kinds of between-data set heterogeneity have on relative model prediction performance of models selected by different information criteria? We address this question in the simulation study, where we simulate data sets according to different heterogeneity regimes and examine RMSE performance of AIC, AIC_C and BIC in terms of prediction quality from models they select.

A useful observation at this point is that, had we data sets from multiple sites or time points, we might naturally tend to fit a hierarchical (or random effect) model; in other words, we model explicitly the variability between sites or times, allowing each to have (in essence) a different model. In practice, however, we often do not have this luxury, being restricted instead to a single data set from a single time or place. In that circumstance, we are not able to estimate the degree of heterogeneity between data sets, but it may have an impact on our modelling; just how much is the topic of the following section.

Simulation study

Some authors have criticized past attempts at conducting simulation studies. While recognizing the importance of simulation, Burnham & Anderson (2002) argued that in practice, studies are often too simplistic or biased against AIC, because:

1. a simple generating model is used, with no tapering, and often only few variables;
2. the set of models considered contains the generating model; and
3. the aim of the simulation study is to select the generating model.

Burnham & Anderson (2002) noted justification for the routine use of AIC via theoretical considerations; however, as the theory is largely based on asymptotic arguments, they also acknowledged that uncovering practical implications for

‘small’ (which technically means ‘not infinite’) data sets requires simulation experiments. Later, Burnham & Anderson (2004) suggested that the ‘proper way to compare AIC- and BIC-based model selection is in terms of achieved performance, especially prediction . . .’. Murtaugh (2009) argued that ‘Simulation, in which the ‘true’ model is known, would seem the only definitive way to compare model-building techniques’. Hence, we adopt this approach here.

We establish a novel simulation framework for linear models with the express aim of studying the effects of different levels and types of unobserved heterogeneity across data sets on model selection. Within the framework we can allow data sets to vary in a number of different ways, and we can use the results to compare the predictive performance of different model selection strategies. We allow any kind and any number (within computational constraints) of effect sizes, variables from the generating model can be omitted from the data set, and other, ‘spurious’ variables can be included. In this way, we avoid favouring (either deliberately or inadvertently) BIC over AIC, a particular concern of Buckland, Burnham & Augustin (1997).

A NOVEL SIMULATION FRAMEWORK

We propose a simulation framework as follows, in which we wish to generate observations from a linear model. For simplicity, we describe the framework in terms of a single observation, avoiding the need to index the observation number. We specify a linear model as at (3), where the response variable y depends on regression intercept α , regression coefficient vector β , covariate vector x , and a residual quantity r ; a similar specification is possible for a GLM. We suppose we have two kinds of covariate: those that are part of the generating model (the ‘ x ’) and those that are not (the ‘ z ’); the latter represent possible spurious variables, not directly affecting the response, but for which we may have data in a real application. An observation of both the x and the z covariates is generated simultaneously via a multivariate normal distribution (MVN) as at (4), to allow for potential correlations between the covariates; we define means (μ_x, μ_z) and a covariance matrix ($\Sigma_{x,z}$) for the MVN. The residual r has a normal distribution centred on 0 with variance σ_r^2 (Table 1).

For further generality, we consider how sampling from different sites or times might affect real ecological or environmental data collection; it is possible that each site might have its

own means for the x and z covariates, its own covariance matrix, and slightly different intercepts and slopes. The latter arises naturally in random effect/random coefficient models; with real data, we might also expect to see different covariate distributions at different sites, for example. We define univariate normal distributions for each element of μ_x and μ_z , expressed in vector notation at (5) and (6), with higher level means (μ'_x, μ'_z) and variances (σ_x^2, σ_z^2). For α and β we give standard random effect distributions, but centred on higher level means α' and β' , with variances as at (8) and (9).

The final part concerns generating and varying the covariance matrix $\Sigma_{x,z}$. We use the method of Hardin, Garcia & Golan (2013) for generating a correlation matrix, which we then convert to $\Sigma'_{x,z}$ using supplied variance terms for x and z . Our implementation of Hardin, Garcia & Golan (2013) takes three parameter inputs: ρ , specifying the mean correlation; and two variability parameters, ε and ε_m , specifying variability about ρ . We explain how this works in practice later in the section headed ‘The link between penalty and predictive performance’.

SIMULATION IN PRACTICE

We provide an R package ICsims (<https://github.com/MarkJBrewer/ICsims/>) for working with our sampling framework. The user specifies sample sizes for each simulation, the number of simulations, and which of the x covariates are to be supplied as data to the model (to avoid the criticism that the generating model is in the candidate set). The code works by simulating the covariates x and z and residuals r , calculating the response y , and fitting all possible models (using the DREDGE function in the MuMIn package, Barton 2015). Then, a second, ‘out-of-sample’ data set is simulated, and the predictions using the fitted models (i.e. the models obtained from the first data set) are compared with the second set of responses; we use root mean square error (RMSE) as our metric of prediction quality. The process just described counts as one simulation in our framework; in the examples below, we run 1000 simulations.

The user supplies values for $\mu'_x, \mu'_z, \alpha', \beta', \sigma_r^2$ and either $\Sigma_{x,z}$ or the parameters for the Hardin, Garcia & Golan (2013) method (i.e. ρ, ε and ε_m) if the covariance matrix is to be generated. If we set the quantities $\sigma_x^2, \sigma_z^2, \sigma_\alpha^2, \sigma_\beta^2$ to zero, then the sampling distributions between the first and second data sets will be identical, and $\mu'_x = \mu_x$, for example. In this way, the generating model can either be constant across samples or highly variable.

For illustration, we consider two patterns for the β coefficients, illustrated by Fig. 1: ‘strong effects’, where we have 10 covariates (the x) in the generating model, all with regression coefficient equal to 5.0 (or, as we see later, 1.0); and ‘tapering effects’, with 21 real effects, starting at value 5.0 but tapering off quickly to very small values. Only the variables whose circle is filled in Fig. 1 will be supplied as data; the empty circles denote omitted variables, not available in the data set. We also define 5 spurious variables in each case (the z), corresponding to coefficients of value 0.0.

Table 1. Technical specifications for simulation framework

$y = \alpha + \beta x + r$	(3)
$\begin{pmatrix} x \\ z \end{pmatrix} \sim MVN \left[\begin{pmatrix} \mu_x \\ \mu_z \end{pmatrix}, \Sigma_{x,z} \right]$	(4)
$\mu_x \sim N(\mu'_x, \sigma_x^2)$	(5)
$\mu_z \sim N(\mu'_z, \sigma_z^2)$	(6)
$r \sim N(0, \sigma_r^2)$	(7)
$\alpha \sim N(\alpha', \sigma_\alpha^2)$	(8)
$\beta \sim MVN(\beta', \sigma_\beta^2)$	(9)

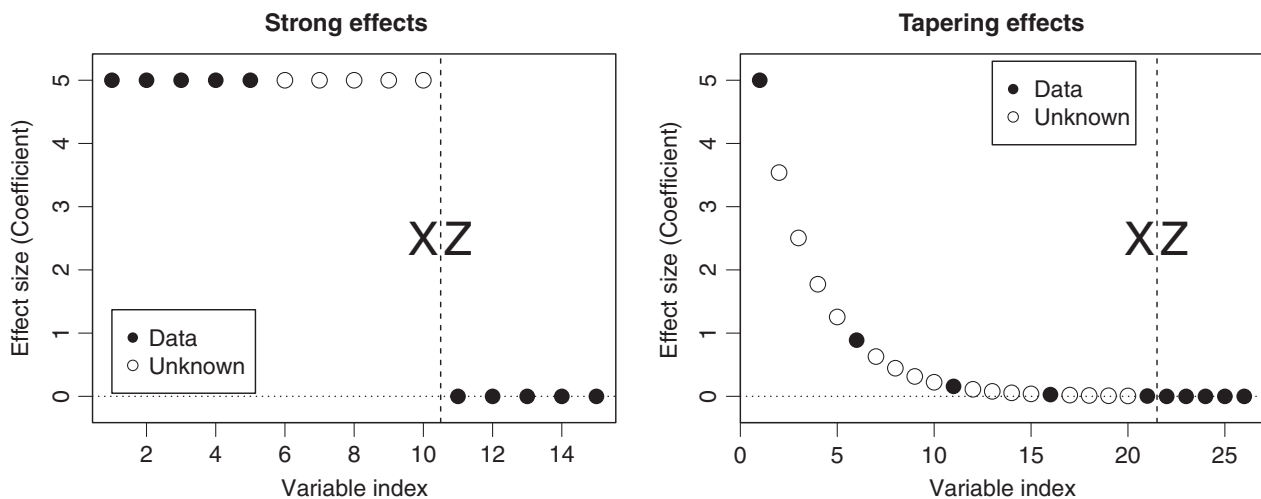


Fig. 1. Illustration of the two effects schemes – strong vs. tapering. The coefficients in the *X* section represent genuine effects, while variables corresponding to the *Z* coefficients are spurious (hence the coefficients have value zero). Only the effects marked by filled circles are made available as data; the empty circles represent coefficients for unknown or omitted variables, used in the generating model but not available as data.

In our simulations, we consider samples sizes from 20 to 10 000, thus ranging from very small to very large ecological data sets.

Results

EXAMPLE 1 – STRONG VS. TAPERING EFFECTS (UNCORRELATED COVARIATES)

We start by considering the *x* and *z* variables generated independently from standard normal distributions, so that: $\mu'_x = \mathbf{0}$; $\mu'_z = \mathbf{0}$; $\sigma_x^2 = \mathbf{0}$; $\sigma_z^2 = \mathbf{0}$; and $\Sigma_{x,z}$ is the identity matrix. We set $\sigma_r^2 = 1$ here and throughout. Figures 2–6 contain two kinds of plot: plots showing the mean RMSE values for each of AIC, BIC and AIC_C, for each sample size (the latter on a log scale) from the simulations; and plots showing the proportion of times a criterion was best in terms of RMSE, where ‘best’ here includes ‘joint best’, as different criteria sometimes select identical models. As a consequence, the proportions in each plot for each sample size need not sum to 1.

Looking first at average RMSEs in the top row of Fig. 2, we see that for the strong effects, there is little to choose between the three metrics, but for the tapering effects, there is a slight advantage for BIC for small-to-moderate sample sizes. This is perhaps a little surprising; we might have expected either AIC or AIC_C to perform better here, given the theory. In the second row of plots, we see that BIC is often the best choice of criterion, even for larger sample sizes. One caveat on these comparisons is that for any given sample size, the differences in average RMSE are very small between the different criteria (except perhaps for the smallest samples); sample size itself is a bigger driver of RMSE value, as would be expected.

The reason for these results should be clear; the difference between AIC and BIC is that AIC will more often select the very weak effects in a taper. For smaller sample sizes, those

effects will be estimated very poorly in relative terms – estimates of tiny positive numbers in this context can go negative, for example; in that case, zero may be a better estimate for the coefficient, which is equivalent to not selecting the variable in the first place. This is why BIC has better predictive performance for the smaller sample sizes; for larger samples, all methods perform equally well, since then, even when spurious variables *are* selected, the coefficients are likely to be tiny and thus the predictions are not affected greatly. AIC_C corrects AIC for smaller sample sizes, but even then there is a slight edge in favour of BIC for all but the smallest.

This finding is in conflict with the earlier statement from Burnham & Anderson (2004) on the optimality and performance of AIC; it turns out that it is not the tapering effects which are favouring AIC (or AIC_C) after all.

EXAMPLE 2 – STRONG VS. TAPERING EFFECTS (CORRELATED COVARIATES)

This example is as Example 1, except that now we introduce correlations between the *x*'s and *z*'s by setting all off-diagonal elements in the covariance matrix $\Sigma_{x,z}$ equal to 0.1. As can be seen in Fig. 3, for the tapering effects this apparently modest degree of correlation is sufficient to tip the scales in favour of AIC and AIC_C, as they now outperform BIC in terms of average RMSE (top row) and frequency of being the ‘best’ model (bottom row). For the strong effects on the other hand, in RMSE terms the correlation does not appear to have affected the relative performance of the metrics greatly, as the between-criteria differences are small, but AIC does appear to provide optimal performance most often (bottom left).

EXAMPLE 3 – LOW VS. HIGH COVARIATE CORRELATION HETEROGENEITY

We now extend the model from Example 2, this time allowing the correlations between the *x*'s and *z*'s to be generated

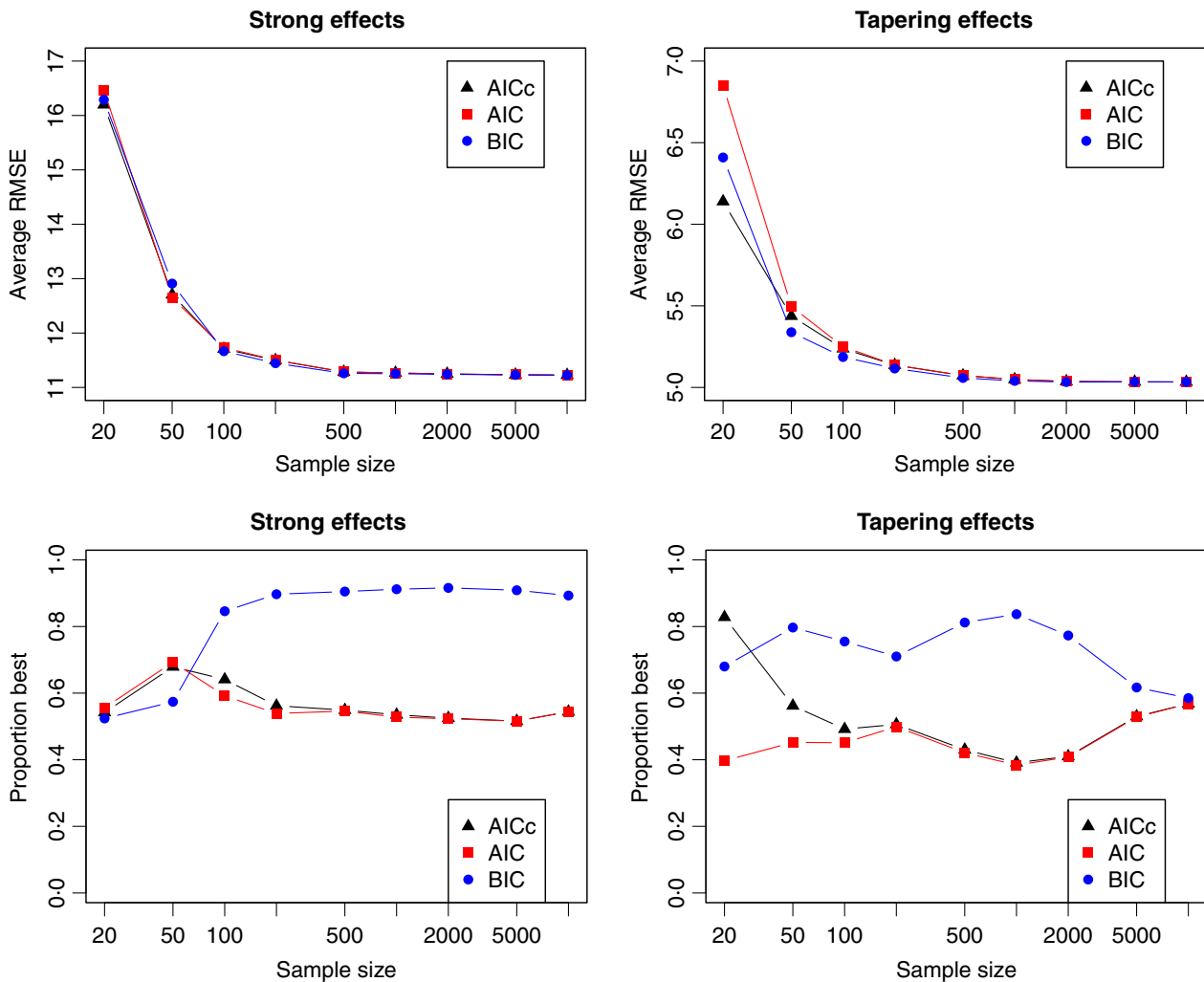


Fig. 2. Top: RMSEs for strong vs. tapering effect models (from Example 1) against sample size. Bottom: proportion of simulations each criterion is best or joint best in terms of RMSE performance.

randomly. We also allow the correlations to change between sampling; we consider both a low and a high heterogeneity regime. For all simulations, we set $\rho = 0.5$, so that the mean pairwise correlation is 0.5, and $\varepsilon = 0.05$, representing a low variability around ρ for the correlations in a generated matrix. We study two different values for the final parameter, ε_m , which represents between-data set variability: first, $\varepsilon_m = 0.1$, suggesting a low degree of heterogeneity, and $\varepsilon_m = 0.9$, allowing for a much higher degree of heterogeneity. In practice, this means that: when $\varepsilon_m = 0.1$, the out-of-sample simulations will be done using a covariate correlation matrix which is very similar to that used to generate the original data which led to the model being tested; and when $\varepsilon_m = 0.9$, the two sets of simulations will use quite different covariate correlation matrices.

Figure 4 shows that with low levels of correlation heterogeneity (rows 1 and 3), the results are very similar to those from the previous example, in that AIC and AICc perform better than BIC. For high correlation heterogeneity (rows 2 and 4): for strong effects, BIC is now more competitive, but perhaps still less favoured; for tapering effects, BIC outperforms AIC

and AICc for a good range of sample sizes. Again, it appears the degree of heterogeneity is important.

EXAMPLE 4 – LOW VS. HIGH RANDOM SLOPE HETEROGENEITY

Here we consider the effect of heterogeneity of the model parameters themselves, in a random effect sense. We consider only the tapering model from Example 2, which includes between-covariate correlations of 0.1, but we now define σ^2 as either a vector of 1's or a vector of 5's, representing low or high between-data set parameter heterogeneity respectively. Figure 5 shows that with low heterogeneity, the selection metrics perform roughly equally, but with high heterogeneity BIC is preferred for moderate sample sizes.

EXAMPLE 5 – WEAK EFFECTS

Our final comparison in this section concerns a weak effect version of the 'strong effect' thread. We take the strong

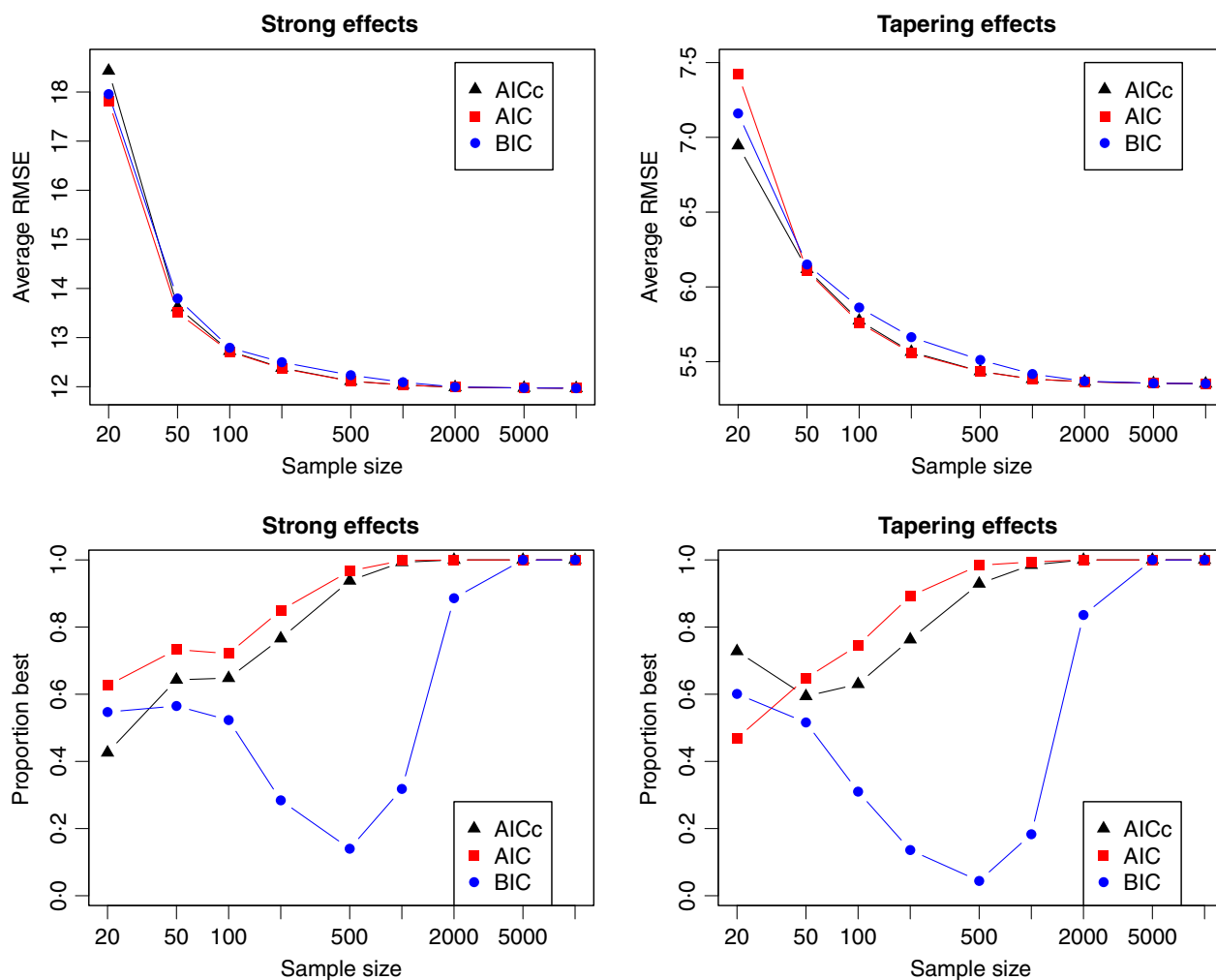


Fig. 3. Top: RMSEs for strong vs. tapering effect models with covariate correlation (from Example 2) against sample size. Bottom: proportion of simulations each criterion is best or joint best in terms of RMSE performance.

effect regime from Example 3 but set the effect sizes (the β') to take the value 1.0 rather than 5.0. Does having a smaller effect size change the relative performance of the information criteria?

The results are in Fig. 6, which should be contrasted with the first column of Fig. 4. The low covariate correlation heterogeneity results (first column) are very similar to those from Example 3 (where AIC and AIC_C outperform BIC). There is a noticeable difference, however, in the results for the high covariate correlation heterogeneity; in Fig. 4, we saw that for the strong effects regime BIC was competitive with but slightly outshone by AIC and AIC_C; with the smaller effect sizes, BIC is now performing substantially better than AIC and AIC_C for sample sizes from 50 up to 2000. So while in Example 3, an increase in covariate correlation heterogeneity did not greatly alter the relative performances of AIC, AIC_C and BIC in the presence of strong effects, if the effects are much weaker then BIC does gain an advantage – and for similar reasons given for the advantage in the tapering results of Example 1.

SUMMARY OF RESULTS

According to our simulations, the key aspect of any data set or real data situation with respect to a choice of information criterion is clearly heterogeneity, relative to effect size. With only large effects, the choice of metric is unimportant. With small effects and moderate sample sizes, and accepting that with almost all data sets in ecology there will be some form of non-trivial correlation among candidate explanatory variables, BIC provided lower prediction error than AIC or AIC_C if either the correlations or the effect sizes themselves are highly variable between data sets. It is this heterogeneity which causes the assumptions behind the derivation of AIC's supposed optimality to fail.

We stress that the simulations from our framework can account for between-data set heterogeneity, but the modelling process does not. As noted earlier, often in practice we do not have replication – we may only have data from one time or place. It is clear from the results above that data set heterogeneity is important, but earlier work – including Burnham &

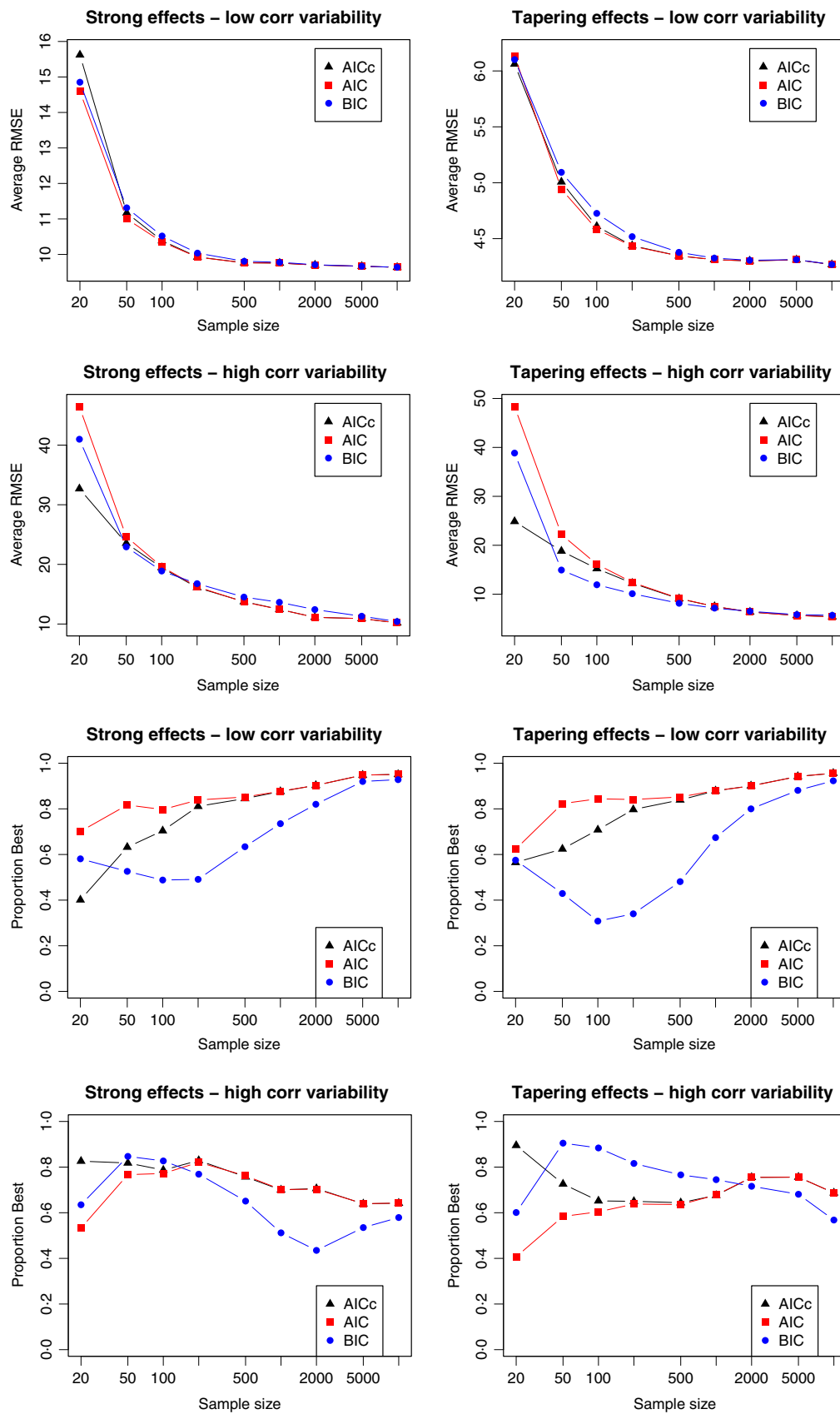


Fig. 4. Top two rows: RMSEs for strong vs. tapering effect models comparing low and high covariate correlation heterogeneity (from Example 3) against sample size. Bottom two rows: proportion of simulations each criterion is best or joint best in terms of RMSE performance.

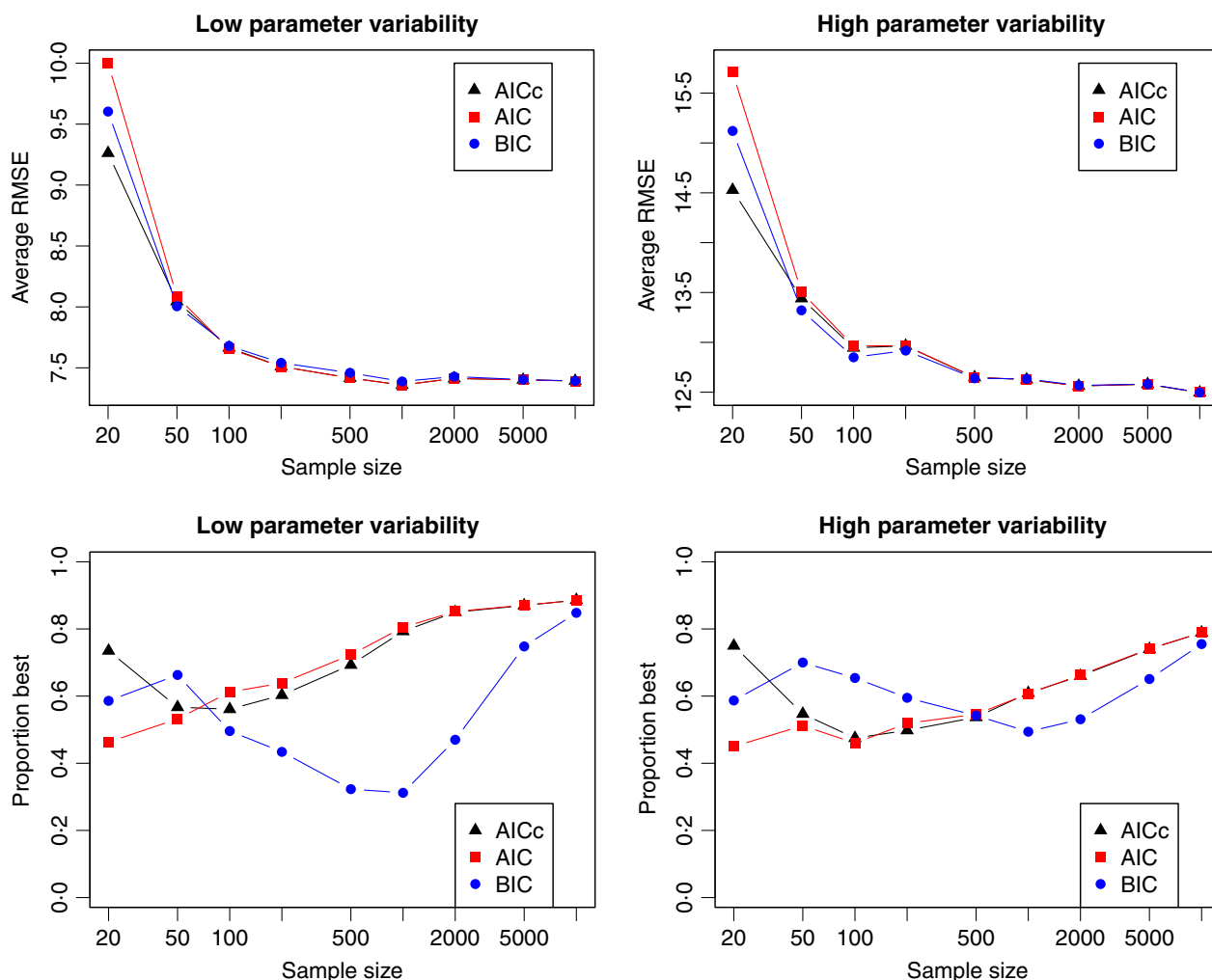


Fig. 5. Top: RMSEs for low vs. high effect heterogeneity (from Example 4) against sample size, tapering model. Bottom: proportion of simulations each criterion is best or joint best in terms of RMSE performance.

Anderson (2002) – did not allow for it. This is why our findings with respect to BIC in terms of selecting models for prediction are novel.

Other than making a general statement about variability, it would appear difficult to forecast with any certainty which information criterion will perform best in any particular modelling exercise. In fact, given that there is no reason to believe that BIC is optimal any more than AIC, we could broaden the question as to what *level of penalty* is appropriate for any analysis. We address these issues in the following section.

The link between penalty and predictive performance

Given that we have seen that no information criterion is necessarily optimal for prediction in practice for all data sets, we would like to have a way of estimating what the optimal penalty might be. Recall from (1) and (2) that the penalty terms are multiples of p , the number of model parameters. In AIC's case the multiplier is 2, and for BIC it is $\log(n)$, the log of the sample size. Here we investigate

briefly, by means of our simulation framework, the relationship between penalty and prediction quality. We consider only linear multiples of p for simplicity, but note that more complex functions would be possible.

We consider two simple simulation schemes, both with the same covariate structure but with different generating models:

- Model 1: $\alpha' = 3, \beta' = (4, -2, 0.01)'$
- Model 2: $\alpha' = 3, \beta' = (0.5, -0.1, 0.01)'$

The covariate simulations are defined by $\mu'_x = (5, 2, 3)$, $\mu'_z = 7$ and

$$\Sigma_{x,z} = \begin{pmatrix} 1 & 0.6 & 0.2 & 0.3 \\ 0.6 & 1 & 0.4 & -0.1 \\ 0.2 & 0.4 & 1 & 0.1 \\ 0.3 & -0.1 & 0.1 & 1 \end{pmatrix}.$$

We set all of σ_x^2 , σ_z^2 , σ_α^2 , σ_β^2 to zero. Thus Model 1 has two strong effects and one very weak, and Model 2 has two weak effects and one very weak. There is a 'spurious' variable included in the data set for analysis in both cases, but which is correlated with variables in the generating model, and so in practice is indistinguishable from a gen-

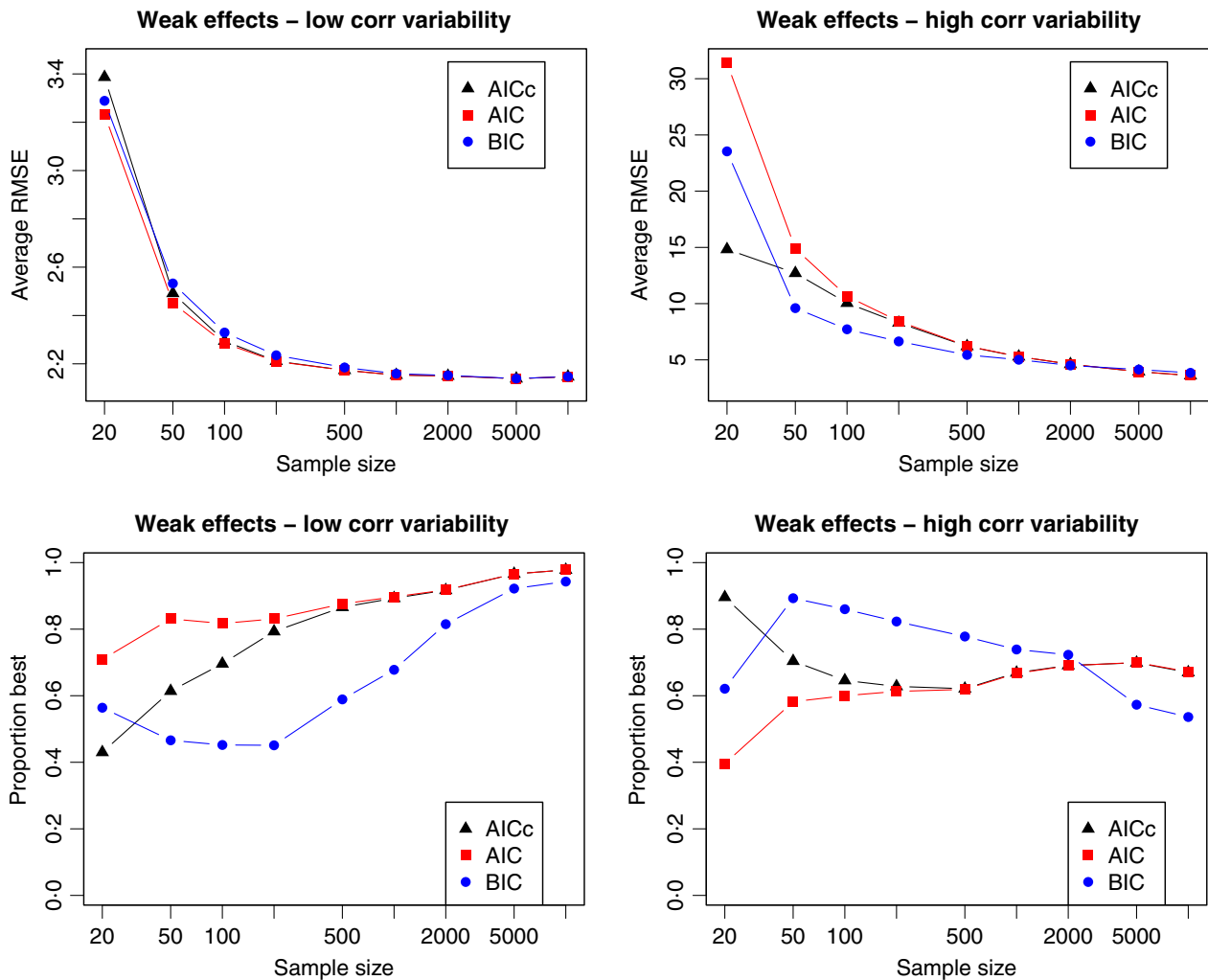


Fig. 6. Top: RMSEs for low vs. high covariate correlation heterogeneity (from Example 5) against sample size, strong effect model. Bottom: proportion of simulations each criterion is best or joint best in terms of RMSE performance.

uine effect. We use the ICSIMS package to obtain the full set of fitted models and associated out-of-sample RMSE scores for each simulation – we run 2000 here, with sample size 100. Then we work out what the average RMSE score would have been, across the simulations, if different penalties had been applied as multiples of p , noting that a value of 2 corresponds to AIC, and $\log(100) \approx 4.6$ corresponds to BIC. AIC_C is not included here as it does not operate as a set multiple of p .

In Fig. 7 we show the average RMSE obtained for two ranges of penalties for each model: (0,100) and (0,25). For Model 1, we see that the average RMSE has a low plateau between 15 and 80. This represents the levels of penalty which are sufficient to ensure selection of the strong effects (i.e. variables x_1 and x_2) but not the weak effect (variable x_3) or the ‘spurious’ effect (x_4). Lower penalties include either x_3 or x_4 , and with $n = 100$, the estimation of x_3 ’s coefficient is imprecise enough to reduce out-of-sample prediction accuracy. When the penalty increases beyond 80 (as a multiplier of p) models with only one of x_1 or x_2 are selected (increasingly often as the penalty increases), and prediction accuracy declines accord-

ingly. Note that neither AIC nor BIC are ‘optimal’ in this particular case; the evidence suggests BIC would be a better choice, but a penalty of 15.1 is the smallest value leading to the minimum average RMSE.

For Model 2, the results are quite different; this time there is a high plateau, which corresponds to only the strongest (but weak) effect (x_1) being selected. There is no penalty which sees the next largest effect x_2 selected consistently, and in fact increasing penalty values from zero just causes x_2 to be selected less and less. On that basis, the ‘optimal’ penalty is close to zero, and AIC is preferable to BIC.

We are not aware of previous work studying the relationship between likelihood penalty and prediction quality in the way we have in this section, but results from this simple analysis are striking – the relationship is a complex one, depending as it does on the precise nature of the relative sizes of effects in a generating model. We did not complicate the issue here by introducing any form of between-data set heterogeneity; we will do that in the next section, where we consider a real data application and how our simulation framework might assist model selection.

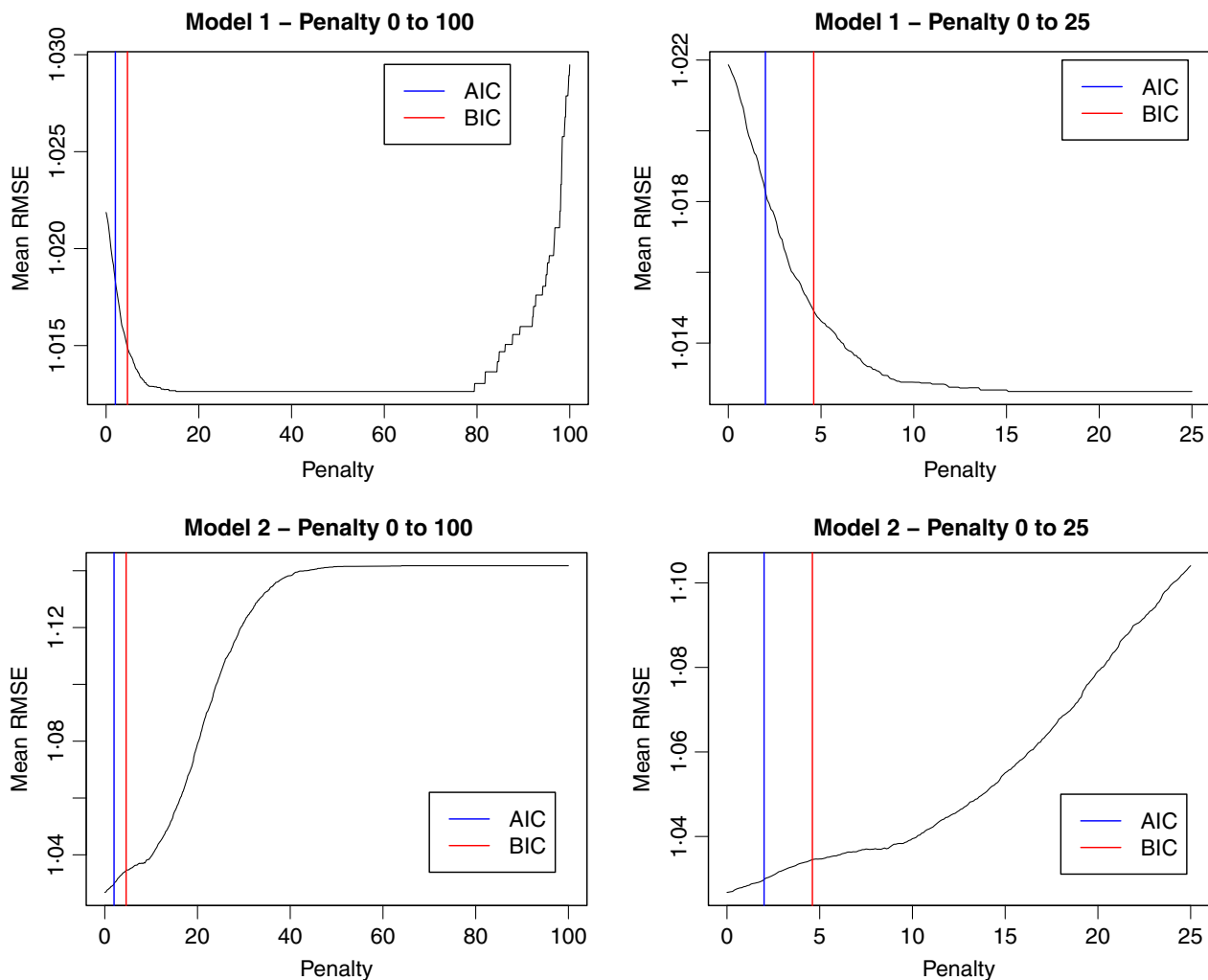


Fig. 7. RMSEs against penalty multiplier on p for simulations exploring the link between penalty and predictive performance. Top: Model 1; Bottom: Model 2.

Application – freshwater pearl mussels

Cooksley *et al.* (2012) studied associations between presence of freshwater pearl mussels and man-made structures in a Scottish river. The response variable was a binary indicator of species presence measured at 50-m sections along the length of the river, and the 31 potential explanatory variables concerned aspects of artificial structures (such as bridges, embankments, treatment works, etc). Here we consider a subset of the original variables, listed in Table 2; for further details on the application, see Cooksley *et al.* (2012), noting that the numbering in Table 2 here does not correspond to the numbering in that paper. The data set contains 1211 observations. We consider models including only linear terms and no interactions for simplicity; our simulation framework considers all $2^9 = 512$ models.

The models fitted are now GLMs, so this requires a slight modification to the simulation framework earlier. The regression equation at (3) is now written in terms of a linear predictor

Table 2. Explanatory variables in freshwater pearl mussel data set

x_1	Number of tributaries within five sections upstream
x_2	Protection on survey bank
x_3	Presence of bridge
x_4	Number of sections upstream to next bridge
x_5	Number of bridges within five sections upstream
x_6	Number of occurrences of dredging within five sections upstream
x_7	Presence of waste water treatment works
x_8	Number of artificial feature categories present
x_9	Deposition zone

η , so that

$$\eta = \alpha + \beta x + r$$

and where the response variable is related to η via the link function. With a binary response, we assume binomial errors and a logit link function. For the simulations, we generate values for the linear predictor η and then convert this to a probability using the (inverse) link function; the final step is to generate the actual response variable from a binomial dis-

tribution using the generated probability and a number of trials equal to 1 (as we are simulating a binary response).

So how can we put a real data set within our sampling framework? Given a single set of data from one time/place, there are a number of potential inputs we need to think about:

- **Effect sizes.** Defining α' and β' . With a real data application, we do not have a set of known parameters from a true generating model. We can, however, use the estimates derived from fitting an appropriate model (such as the full model, which contains all of the explanatory variables) to the observed data set as a basis for simulation.
- **Properties of covariate distributions.** It should be straightforward to obtain values for μ_x , μ_z and $\Sigma_{x,z}$, as these can be calculated from the data set.
- **Residual variance.** For linear models, the residual variance σ_r^2 could be estimated from fitted models, as for effect sizes.
- **Heterogeneity settings.** These are the variance parameters on the effects (σ_α^2 and σ_β^2) and the covariates (σ_x^2 and σ_z^2) and the degree to which the correlation matrix between the x 's and z 's is allowed to vary. There may be information to help us, however; for example, we may have covariate data available from different sites or locations, even if we do not have response data. In that case, we could use these extra data sets to inform our choice of σ_x^2 , σ_z^2 and so on. The effect size variances are a little harder; we may simply need to perform a sensitivity analysis, trying different values for σ_α^2 and σ_β^2 and assessing what impact this has on the prediction performance of different penalties or information criteria. Any external knowledge on the stability or otherwise of effects across data sets would be useful in guiding choices for σ_α^2 and σ_β^2 .

For this example, we obtain effect sizes from the full GLM with all nine explanatory variables (there are no z 's) and consider four different heterogeneity regimes within our simulation framework:

1. **Static sampling.** All of the heterogeneity settings are at zero.
2. **Variable covariate means.** As 1., but with $\sigma_x^2 = 5$, for this data set representing a very high level of covariate heterogeneity.
3. **Low effect variability.** As 1., but with σ_β^2 equal to a vector of values 0.05, representing a **low** degree of effect heterogeneity.
4. **High effect variability.** As 1., but with σ_α^2 equal to a vector of values 0.10, representing a **high** degree of effect heterogeneity.

Results are shown in Fig. 8. For the 'static sampling' case, small penalties are preferred, and in terms of RMSE, both AIC and BIC seem close to optimal. The same is true of the 'variable covariate means' case; for this example, having unstable covariate means across samples does not appear to alter the shape of the RMSE vs. penalty relationship. With increasing effect variability, however, it seems that increasing penalties achieve optimal RMSE. The effect on model choice for these higher penalties is dramatic; the model selected by AIC includes variables x_1 , x_2 , x_6 , x_8 and x_9 , whereas the optimal penalty for low effect variability (value 15.17) selects the model with x_1 alone, and in the case of high effect variability (optimal penalty 35.2) none of the explanatory variables are selected. This tells us that models are sensitive to relatively small amounts of variability

in the effect sizes; only the effect of variable x_1 is robust under a degree of effect heterogeneity of 0.05 for these data.

Discussion

Submitting papers to ecological journals, we have had personal experience of reviewers *insisting* that AIC be used, preferably in the manner espoused by Burnham & Anderson (2002), when we have not felt it necessary or even appropriate. This 'one size fits all' world view is not supported by the results in this paper; as long ago as 2005, Stephens *et al.* (2005) were calling for 'pluralism', the notion that there might be different, valid ways of performing model selection. We have focussed on the use of information criteria – other aspects (especially the role of hypothesis testing) being dealt with elsewhere (e.g. Murtaugh 2014a,b) – and shown how in practice AIC (or AIC_C) is not necessarily optimal in all cases, especially when there is a high degree of unobserved between-sample heterogeneity. That theoretical optimality only extends to situations where the data-generating mechanisms and relationships are relatively stable, something borne out by our simulation study. Wenger & Olden (2012) found that models transferred poorly from one site to another unless model complexity was constrained; we believe the method in the section exploring the link between penalty and predictive performance and the practical data exploration suggested in the freshwater pearl mussels application could be seen as a 'transferability assessment', recommended by Wenger & Olden (2012).

This study has focussed on whether the claims made about AIC (or AIC_C) held in practice, and hence, our simulation study looked at RMSE exclusively. There are other criteria which could be used for model selection and comparison. For example, in some applications accuracy may be a more important concern than precision, and hence, a measure of bias of predictions might be preferred to RMSE. Rather than just assuming that RMSE is the 'correct' criterion, how to evaluate the quality of predictions should be considered on a case-by-case basis. This may be especially relevant in the case of non-normal responses, where correct allocations to classes or groupings may be more important than simple numerical accuracy or precision. We have left aside considerations of model assessments not related to prediction, for example identifying effects and estimating effect sizes, as claims for AIC are less relevant in such cases.

There are two further issues raised by Burnham & Anderson (2002): first, null hypothesis significance testing (NHST) for model selection is 'often poor'; and secondly, rather than focussing on a single 'best' model, inference should be made on the basis of multiple models, ranked and/or weighted using calculated AIC values. We discuss these issues only briefly here.

INFORMATION CRITERIA VS. HYPOTHESIS TESTING

We are aware, as are others (for example, Whittingham *et al.* 2006), of problems associated with *stepwise regression* using traditional F -tests (or likelihood ratio tests, LRTs) for choosing between models. Freedman (1983) showed how and

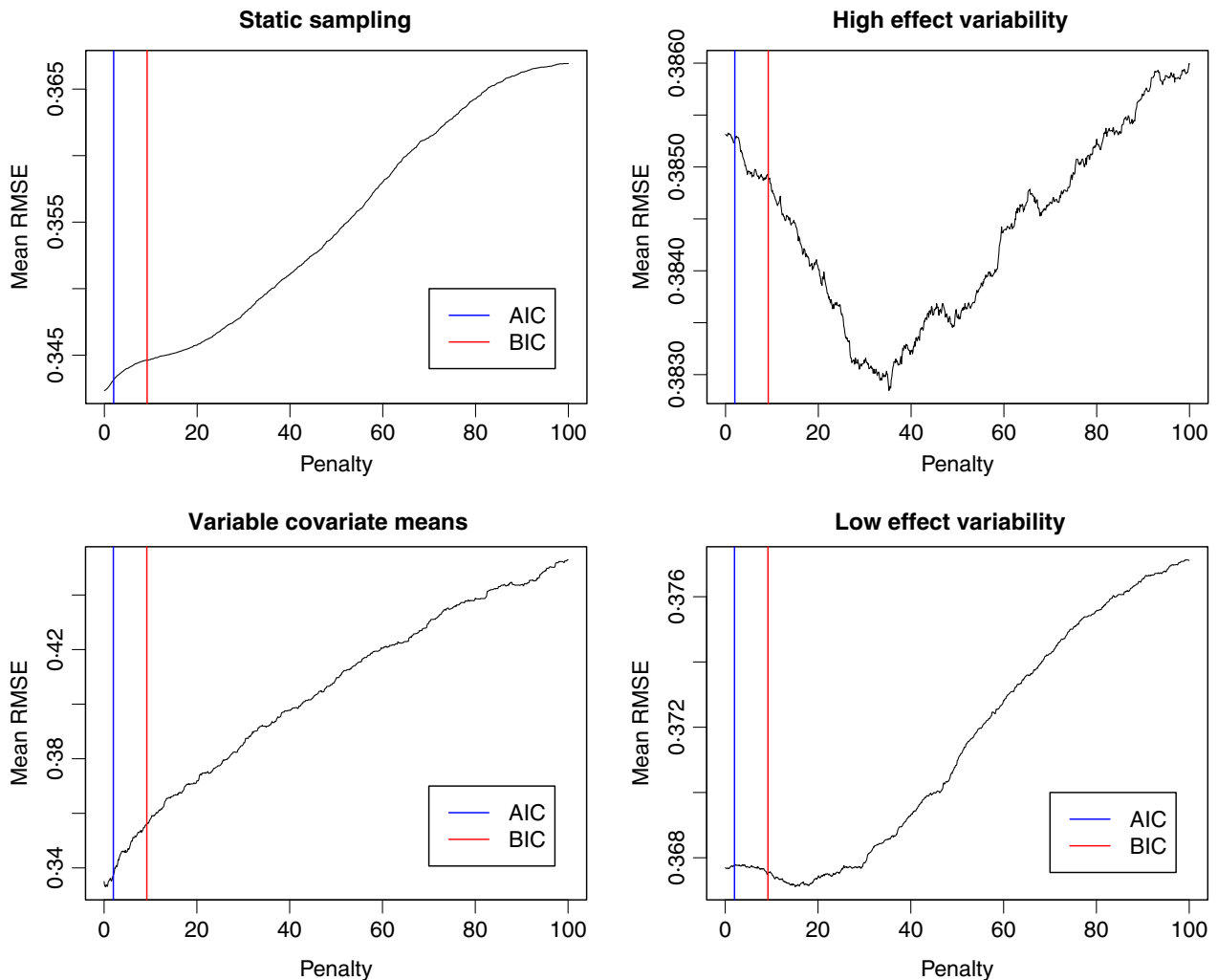


Fig. 8. RMSEs against penalty multiplier on p for freshwater pearl mussel application. Clockwise from top left: static sampling example; high effect variability example; low effect variability example; and example with variable covariate means.

why P -values of final models are not valid for inference – they are conditional results from multiple testing. While we share the concerns about stepwise procedures, we argue that NHST is useful when comparing a small set of models (in a pairwise fashion) when examining a particular hypothesis. As Murtaugh (2014a) stressed, AIC and P -values are directly related for given sample sizes (see fig. 2 in Murtaugh 2014a, for example); also, note that when comparing two models using AIC, one looks at ΔAIC :

$$\Delta\text{AIC} = -2\log \ell(\hat{\theta}_1) + 2\log \ell(\hat{\theta}_2) - 2(p_2 - p_1)$$

while a LRT compares the following

$$-2\log \ell(\hat{\theta}_1) + 2\log \ell(\hat{\theta}_2)$$

to a $\chi^2_{p_2-p_1}$ distribution. These expressions differ only in a term equal to twice the difference in the numbers of parameters in the two models; when comparing two models, therefore, the two approaches are very similar. Murtaugh (2014a) notes that

in a comparison of two models which differ by one parameter, preferring one model over another on the basis of AIC is equivalent to using a P -value significance level of 15.7% in a LRT, and so AIC is more liberal in terms of including variables. Burnham & Anderson (2002, §6.9.3) give an example which they claim illustrates the theoretical finding that when comparing two models, AIC has a ‘sound theoretical basis’ whereas LRTs do not. They consider a set of models, all with the same AIC, which differ by j parameters from a null model. From the equations earlier, we can see that the LRT statistics are equal to $2j$; as j increases from 1 to 30, the LRTs produce P -values which go from 0.157 to 0.001 – that is, they change, and produce different inferences for different j , unlike AIC. It is the constancy of AIC here which supposedly provides the ‘sound’ basis; however, this reasoning is flawed, as the simple explanation is that the two approaches are testing very different things. Consider when $j = 30$, so that we are comparing a null model with another containing an extra 30 explanatory variables; AIC is asking ‘is it worth including *all* of the 30 variables?’ whereas the LRT is asking ‘is it worth including *any* of the 30 variables?’ Another way of looking at this is that AIC requires

an improvement in log-likelihood sufficient to justify the inclusion of all 30 variables, whereas the LRT in theory needs only an improvement sufficient to justify even a single extra variable. Either way, these are fundamentally different questions, so of course the answers are different.

MULTIMODEL INFERENCE

We agree that model selection should not, ideally, present only a single model as ‘best’; there are likely to be competing models, none of which can be concluded as categorically ‘better’ than the rest. Rather, the set of models should be presented and compared, and the model uncertainty noted. However, the AIC-based approach to multimodel inference put forward by Burnham & Anderson (2002) has recently been criticized by Cade (2015), who noted, for example, that AIC is a statistic relating to models, so that any form of model averaging done at the level of the explanatory variables will be dependent on the set of candidate models, and ignores the effect of multicollinearity; Murray & Conner (2009) found that AIC was a poor indicator of a variable’s ‘importance’. In addition, the use of AIC to calculate model ‘weights’ (often loosely, but incorrectly, referred to as ‘probabilities’, which strictly would require a Bayesian approach) suffers from the ‘add a spurious variable’ problem; if you take the best model by AIC, and add a randomly generated explanatory variable to form a new model, the ΔAIC will necessarily be less than 2 (as the log-likelihood can only increase, and p increases by 1). Thus, any model which is the best model plus *any* other variable will automatically have a relatively large weight. Burnham & Anderson (2002) discuss this issue in their text box on p. 131; they state in this situation we should see whether the two models have ‘essentially the same’ log-likelihood values, to assess whether the addition of the extra variable was warranted. But how are we to judge what is ‘essentially the same’? Ironically, the standard technique in statistics for determining whether the difference in log-likelihood between two models is ‘enough’ to justify the larger model is a LRT, the very thing Burnham & Anderson (2002) are trying to avoid.

The ‘add a spurious variable’ problem was addressed by Buckland, Burnham & Augustin (1997, p. 607). Consider two competing models which differ only by inclusion/exclusion of one, spurious, variable. AIC weights will weight both models highly, in effect (almost) doubling the weight for the smaller model. A nonparametric bootstrap method, which involves selecting a single ‘best’ model from each of a number of bootstrap resamples, will only ever select one of the models for each resample; this helps avoid the ‘double weighting’ issue.

There have been a number of other studies looking at the performance of different model selection strategies. Murtaugh (1998) and Murtaugh (2009) both concluded after simulation studies that the choice of model selection approach seems to have little bearing on results, either in terms of identifying important variables or in terms of prediction quality – with the possible exception that regression tree methods seem to perform relatively poorly. We have not discussed Bayesian model selection approaches, but for a recent review see Hooten

& Hobbs (2015). Further developments are being made, and new forms of information criteria have been proposed. The Watanabe–Akaike (or Widely Applicable) Information Criterion (WAIC; Watanabe 2010a,b, 2013) is an attempt from Bayesian statistics to emulate a cross-validation analysis; while preferred to DIC, a direct leave-one-out criterion is now recommended (Vehtari, Gelman & Gabry 2015). There have been attempts to generalize or modify AIC and BIC in various ways; a particularly interesting example is found in Lv & Liu (2014), who decompose one of their criteria (GBIC_p) into three parts: goodness-of-fit; model complexity; and model misspecification. Most information criteria address only the first two parts, and as Lv & Liu (2014) found, estimating the third part is difficult – it requires knowledge of the true model, generally unknown; the authors acknowledge that further research is needed in this area.

APPLICABILITY OF FINDINGS

As a final comment, we note that throughout this paper we have referred to data sets recorded at a single location or at a single point in time. This will be a very common scenario, but more generally, our findings hold in any situation where all of the observations have come from a single sampling unit (whether that is site, year, region, individual, species, etc.) but where there is an interest in generalizing beyond that unit; while this situation may be regarded as undesirable, in practice it may be unavoidable. Our findings are also relevant where there is only limited replication, insufficient to allow the variability between sampling units to be estimated as part of the modelling (via, e.g., random effect or hierarchical models).

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Data accessibility

We provide the R package ICsims for conducting the type of simulations in this paper, hosted on GitHub, at <https://github.com/MarkJBrewer/ICsims>. We also make available the set of code and simulated data to reproduce the specific results in this paper on FigShare, at <https://figshare.com/s/9f6909b8c5fe33dcef55>. The one exception to this is the data from the real application; we are unable to share these data for licensing reasons.

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